Whats New (SCL 2009 SPICE Library vs SCL 2020 SPICE Library)

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Part-A : Comparison of new & old pdk

Resistors

1) An additional parameter has been added to 'deltaW' parameter to account for change in width of resistor due to lithography effects. The effective width expression is changed. Below is example for NMOPO resistor.

Old pdk	New pdk
.paramdw = 0.031e-6	.paramdw = '0.031e-6+dw_lytho_2t'

2) Voltage coefficients have been provided as ZERO in new pdk for all poly resistors. Old pdk has finite values of voltage coefficients for all resistors. Below is example for NMOPO resistor.

Old pdk	New pdk
.param tc1 = -1251e-6 .param tc2 = 2.04e-6 .param vc1 = 2.10e-04 .param vc2 = -3.65e-04	.param tc1 = -1251e-6 .param tc2 = 2.04e-6 .param vc1 = 0 .param vc2 = 0

However, it has been decided that finite voltage coefficients shall be used. Please use the values as given in the SPICE file provided to you.

3) Temperature coefficients are similar.

4) Mismatch coefficients are similar.

5) The SPICE limits for sheet resistance are same in both PDKs.

MIM Capacitors

1) Temperature coefficients are similar.

2) Voltage coefficients have been changed.

3) Mismatch coefficients have been changed.

Old pdk	New pdk
.paramapelg_sqr=1.91e-4 .parambpelg_sqr=2.4e-6	.paramapelg_sqr='0.5*7.05e-5' .parambpelg_sqr='1.0*1.7e-6'

As decided (on 20 Oct 2022), the old values of mismatch parameters shall be used till further revision by the Device Engineer. Please use the values as given in the SPICE file provided to you.

4) 2 diodes across the MIM terminals have been added in both directions, to account for leakage current. Breakdown voltage of diodes used is 58V. The model of diode used contains expressions for tunneling current and breakdown.

Old pdk	New pdk
c1 n1 n2 '<capacitace expression>'	c1 n1 n2 '<capacitace expression>' D1 n2 n1 dcmim area=2 D2 n1 n2 dcmim area=2 .model dcmim d level=1 eg=0.01 is=1e-15 jtun=itun_dntun='53*t_ntun' bv=36 ibv=1e-7 nbv=0.1 n=280 rs=300

Accumulation Capacitors

1) Capacitance expression are same in both pdk.

2) 2 diodes across the Accumulation capacitor terminals have been added in both directions, to account for leakage current.

Old Pdk	New Pdk
Cg g x C='<capacitance expression>' Rc x b R='R0_nwell*(lreal/wreal)'	Cg g x C='<capacitance expression>' Rc x b R='R0_nwell*(lreal/wreal)' D1 b g dcnwp area=A_ratio D2 g b dcnwn area=A_ratio .model dcnwn d level=1 is='1e-15*ik_fac' n='4.5*t_n' ikf='5e-13*ik_fac' rs='6e09*rs_fac' trs=-5e-3 eg=0.1 xti=1.0 .model dcnwp d level=1 is='1e-15*ik_fac' n='7.5*t_n' ikf='1e-13*ik_fac' rs='3e12*rs_fac' trs=-5e-3 eg=0.1 xti=1.0

Diodes

1. Diode models (for 1.8V and 3.3V diodes) are same in old and new pdk , except for one type- 'Isolated Pwell/Deep Nwell diode'.

2. In new pdk, a breakdown voltage parameter (bv=13.8) has been additionally specified for 'Isolated Pwell/Deep Nwell diode' in the new pdk. Rest everything is similar.

BJTs

1) Old pdk has direct model for BJTs , but subcircuit model for BJT has been used in new pdk

Old pdk	New pdk
.model vd10 pnp level=1 + is = 7.047e-18 + bf = '3.08*bf_vd10' + nf = '0.9802*nf_vd10' + vaf = '214*vaf_vd10' + ikf = 0.002198	.subckt vd10 c b e m=1 .param + area=1 areab=1 areac=1 + vpnpismatch = agauss(0,1,1) + vpnpisematch = agauss(0,1,1) + is_mm = '(1-1.8375e-03*vpnpismatch/sqrt(m))' + ise_mm = '(1-1.5500e-01*vpnpisematch/sqrt(m))' qvd10 c b e vpd10 area=area areab=areabareac=areac .flat qvd10 .model vpd10 pnp level=1 + is = '7.047e-18 *is_mm' + bf = '0.8790*3.08*bf_vd10' + nf = '0.9802*nf_vd10' + vaf = '214*vaf_vd10' .

MOSFETs

For MOSFETs, the values of parameters have been changed in new pdk. However, this has been checked by simulations of input and output characteristics – found similar with little variations.

1) Subcircuit approach has been used in new PDK for MOSFETs and statistical model option has been provided for 1.8V MOSFETs (both p and n type).

Old pdk	New pdk
.lib np18mos .model n18.1 nmos (+ lmin = 1.65e-007 lmax = 5e-007 wmin = 2.05e-007 wmax = 5e-007 + level = 49 version = 3.24 + vth0 = '0.378117 + delta_n_vth0_18' lvth0 = -1.47403e-008 wvth0 = 2.66725e-008 pvth0 = -3.56297e-015 + k1 = 0.545245 lk1 = 2.98012e-008 wk1 = -3.37317e-008 pk1 = 2.15531e-015 + k2 = -0.0828741 lk2 = -3.13688e-009 wk2 = 2.2083e-008 pk2 = -2.21616e-015 + u0 = '78.1962 + delta_n_u0_18' lu0 = -7.21406e-007 wu0 = 3.91347e-005 pu0 = -3.05508e-012	.lib np18mos .subckt n18 d g s b l=1u w=1u nrs=1 nrd=1 m=1 as=0 ad=0 ps=0 pd=0 +Aw='log(w*10)/abs(log(w*10))' +myHdif= '0.5*(0.15*(1+Aw)+0.15e-6*(1-Aw))' +aseff = 'int(1-as*1.5e-7/myHdif)*2*w*myHdif+as' +pseff = 'int(1-ps*1.5e-7/myHdif)*(2*w+4*myHdif)+ps' +adeff = 'int(1-ad*1.5e-7/myHdif)*2*w*myHdif+ad' +pdeff = 'int(1-pd*1.5e-7/myHdif)*(2*w+4*myHdif)+pd' .param + delvtn_lv = '4.8518*1e-3/sqrt(2*w*l*m)*0.5*((1+Aw)+1e-6*(1-Aw))' + sigvtn_lv = AGAUSS(0,delvtn_lv,1) m d g s b n18stat w=w l=l delvto=sigvtn_lv as=aseffps=pseff ad=adeffpd=pdeffnrs=nrsnrd=nrd .flat m .model n18stat.1 nmos (+ lmin = '1.65e-007' lmax = '5e-007 + 2*delta_n_lint_18' wmin = '2.05e-007' wmax = '5e-007 + 2*delta_n_wint_18' + level = 49 version = 3.3 + vth0 = '0.378117 + delta_n_vth0_18' lvth0 = -1.47403e-008 wvth0 = 2.66725e-008 pvth0 = -3.56297e-015 + k1 = 0.545245 lk1 = 2.98012e-008 wk1 = -3.37317e-008 pk1 = 2.15531e-015 + k2 = -0.0828741 lk2 = -3.13688e-009 wk2 = 2.2083e-008 pk2 = -2.21616e-015 . . . .

2) Upon simulating the current-voltage characteristics, the differences are insignificant. As an example the results of 1.8V NMOS are provided below

	Parameter	**Old** PDK (SPICE Limits – FF / SS)	**New** PDK(SPICE Limits – FF / SS)	Remarks
LVN 10 x 0.18	Threshold Voltage (Vt)	(321 mV / 522 mV)	(321 mV / 522 mV)	SAME
LVN 10 x 10	Threshold Voltage (Vt)	(316 mV , 342 mV)	(316 mV , 342 mV)	SAME
LVN 10 x 0.18	IdSat (A)	(5.06 mA / 6.84mA)	(5.06 mA / 6.84mA)	SAME
LVN 10 x 10	IdSat (A)	(0.219 mA / 0.246 mA)	(0.219 mA /0.246 mA)	SAME
LVN 10 x 0.18	Ioff (A)	(1.55nA, 7.71pA)	(1.65 nA , 7.79 pA)	**SIMILAR**
LVN 10 x 10	Ioff (A)	(9.13 pA , 6.62 pA)	(9.76 pA , 6.69 pA)	**SIMILAR**

3) Changes in values of parameters have been noticed for noise parameters. However, as per query with TSL, the new values of noise parameters should be used. Also it has been found that the new parameters result in higher noise values or 1-2 orders as compared to old SPICE parameters.

Old pdk	New pdk
+ noia = 7.943E+19 noib = 251188 noic = -2.511E-12	+ noia = 1E+21 noib = 65890 noic = -8.195E-13

4) The value of 'xpart' parameter has been updated from 0.5 to 1 in the new SPICE. Xpart parameter is used to define the charge partitioning ratio between gate-drain and gate-source.

Impact of 'xpart' parameter in SPICE model on Gate-Delay Simulation using Ring Oscillator circuit					
	SPICE Library	**XPART**	**Period (ns)**	**Gate-Delay(ps)**	
1	NEW	1	31	28.65	~2% fast
2	NEW	0.5	31.6	29.205	
3	OLD	0.5	31.6	29.20	
4	OLD	1	31	28.650	~2% fast

Changing of 'xpart' parameter from 0.5 to 1, deceases the simulated gate-delay. But the measured results show that gate-delay of the SCL fabricated circuits is on the lower side. Hence xpart value of 0.5 shall be retained in the SPICE file.

Part-B :

New Devices added in the SPICE Library 2020:

1)	5V Mosfets	n5v,p5v	
2)	5V Diodes	dn50,dp50	
3)	5V Native Transistor	nat5v	
4)	5V Accumulation Capacitor	ch53t	
5)	Metal Fringe Capacitors	usgcm52t, usgcm53t	|6Metals |
		usgcm42t, usgcm43t	|5Metals |
		usgcm32t, usgcm33t	|4Metals |
6)	HIPO-2k Resistor	mr22t	